

Aadit Gupta

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EDUCATION

University of Toronto

Bachelor of Applied Science and Engineering in Computer Engineering + PEY Co-op

Toronto, ON

Sep. 2024 – Apr. 2029 (expected)

- Relevant Courses: Engineering Strategies and Practices; Electrical Fundamentals; Calculus III; Computer Fundamentals (C/C++); Digital Systems (Verilog/FPGA Design)

EXPERIENCES

Research Assistant - CAELUS Project | University of Strathclyde

Jun. 2026 – Present

- Built an ML system to help optimize emergency drone deployment for cardiac-arrest response across Glasgow, extending CAELUS, a peer-reviewed framework used by NHS Greater Glasgow and Clyde, to improve on its coverage and routing methodology.
- Engineered a demographic-aware demand model using Poisson NLL loss with spatial k-fold cross-validation across NRS single-year-of-age, SIMD health domain, and 35,000+ row weather datasets.
- Designed a multi-objective NSGA-II station placement optimizer, introducing four improvements over the original paper: wind-geometry-aware coverage matrices, robust coverage estimation across 100 wind scenarios, seasonal demand weighting, and an explicit Pareto front replacing the paper's fixed scalarization.
- Developed a population-density-aware cost surface with A* routing for real-time drone path planning, paired with a heteroscedastic neural network surrogate (FlightTimeNet) for flight-time estimation with uncertainty quantification.
- Presented weekly pipeline progress and technical results to supervisors via structured presentations.

Product Management Intern | Andersen UAE

Jun. 2025 – Aug. 2025

- Developed an AI-powered E-Invoicing Assistant that processed client tax queries in under 5 seconds.
- Built and deployed a RESTful API with FastAPI enabling seamless integration into client systems.
- Cleaned and Analyzed 10000+ data entries using Pandas and stored them in a ChromaDB Vector Database.
- Optimized responses using retrieval + caching techniques from the Database decreasing response times by 40%.
- Engineered response pipelines using OpenAI's GPT-4 LLM through the OpenAI API reducing latency by 1.3 seconds.

CO-CURRICULAR ACTIVITIES & PROJECTS

University of Toronto Autonomous Scale Racing | Hardware Team Member

Sep. 2025 – Present

- Wrote code in C++ that enabled autonomous communication between an Arduino and an Nvidia Jetson.
- Designed a custom part of a PCB on Altium that allowed communication between the Arduino Nano and the Jetson.
- Debugged UART-based communication by addressing timing issues and data size constraint of 32 Bytes on the Jetson.
- Authored hardware-software interface documentation adopted by both the 4-person hardware and software subteams, cutting onboarding time for new members by 30% and reducing integration bugs during testing.
- Created branches on Git to test different optimizations for communication optimizing development exponentially.

University of Toronto Aerospace Team | Software Team Member

Sep. 2025 – Present

- Developed a camera-based target localization system using Python and OpenCV for autonomous missions.
- Implemented minimum-jerk trajectory planning in MATLAB/Python, focusing on smooth and stable flight paths.
- Containerized workflows using Docker to ensure reproducible development environments and consistent testing.
- Collaborated with the hardware subteam to integrate the systems, enabling 4 successful autonomous mission tests.

Camera Strap Redesign – Engineering Strategies | Communication Liaison

Jan. 2025 – Apr. 2025

- Worked with a client to design a solution to improve camera straps by making it 42% more ergonomic.
- Served as sole point of contact between the client and a 5-person team, delivering 2 status reports and ensuring zero missed deadlines across the project.
- Created a Gantt Chart and 2 Status Reports to keep track of the team's progress.
- Prototyped 3 potential designs, iterating based on client and user feedback to achieve an optimized solution.

Streetview Monopoly Game Project

Jan. 2026 – Mar. 2026

- Built a full-stack multiplayer game leveraging Google Streetview for location-based gameplay.
- Developed server-side logic for turn-based state management, player actions, and game rule enforcement.
- Integrated Google Maps API with environment-based configuration for secure API key handling.
- Contributed to feature development in a collaborative Git-based workflow with bi-weekly sprint discussions.

FPGA Design Project

Sep. 2025 – Nov. 2025

- Designed a rhythm game using the DE1-SoC Board, which incorporated a VGA Display, Keyboard and Speakers.
- Implemented FSMs, Shift Registers and Counters in Verilog to implement game logic and output graphics on the VGA.
- Created ModelSim testbenches to simulate and test the Verilog code which assisted in debugging issues.

TECHNICAL AND PROFESSIONAL SKILLS

Coding Languages: Python, C/C++, TypeScript, SQL

Machine Learning & Data Science: NumPy, Pandas, scikit-learn, PyTorch, NSGA-II / multi-objective optimization, MCMC, spatial statistics

Frameworks & Libraries: React, Node.js, Django, Tailwind, FastAPI, SQLite, ChromaDB, OpenCV

Tools & Technologies: Git, Docker, MATLAB, LTSpice, Altium